

Project Design Phase Solution Architecture

Date	30 June 2026
Roll Number	23A51A0535
Project Name	Library Book Request
Maximum Marks	4 Marks

Solution Architecture:

The solution architecture for this project establishes a robust, automated digital framework on the ServiceNow platform, bridging the gap between manual campus administrative problems and cloud-based automation solutions.

Its architecture achieves the following goals:

- **Find the Best Tech Solution:** Uses ServiceNow to replace slow, manual library logging with an automated, paperless database system.
- **Structure and Behavior:** Uses client-side validation scripts to stop past-date entry errors and a secure backend workspace for the librarian.
- **Features and Requirements:** Maps out core components like unique request tracking numbers (LBRXXXXXX), data validation rules, and automated email alerts.
- **Specifications for Delivery:** Sets up a clean custom table database structure (u_library_book_request) so the system is easy for IT to manage and maintain.

Example - Solution Architecture Diagram:

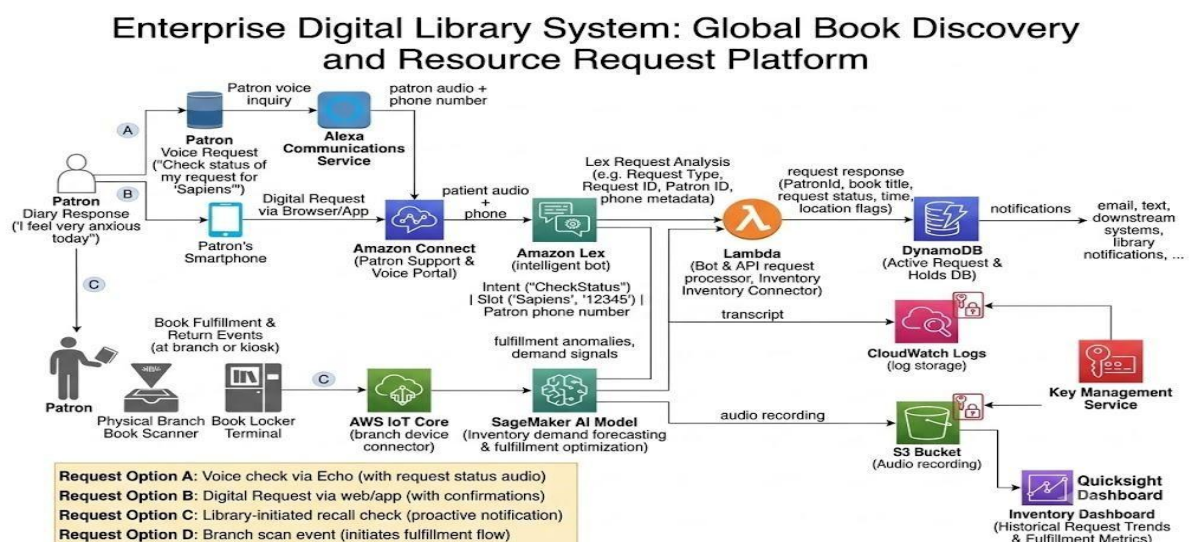


Figure 1: Architecture and data flow of the Enterprise Digital Library request platform 

Reference: <https://aws.amazon.com/blogs/industries/voice-applications-in-clinicalresearchpowered-by-ai-on-aws-part-1-architecture-and-design-considerations/>

